

BI MCQ Quick Revision - All Topics

BI — MCQ Quick Revision (All Topics)

1. Business Intelligence Basics

- BI = turning raw data into actionable insight for decision-making using processes + tools.
- **DIKW order:** Data → Information → Knowledge → Wisdom.
- **Analytics order (imp):** Descriptive → Diagnostic → Predictive → Prescriptive.
- Descriptive = what happened; Diagnostic = why; Predictive = what will happen; Prescriptive = what to do.
- **BI evolution order:** Data Processing → MIS → DSS → EIS → Data Warehousing → BI → Big Data/AI Analytics.
- **ETL** = Extract, Transform, Load.
- **OLAP operations:** Roll-up = summarize/aggregate; Drill-down = go to detail; Slice = one dimension filter; Dice = sub-cube of 2+ dimensions; Pivot = rotate axes.
- OLAP is for analysis (multidimensional); OLTP is for daily transactions (fast, simple queries) — commonly confused pair.
- Metadata = “data about data.”
- Key BI challenge often tested: **data quality/integration issues**, not lack of tools.

2. Information Gathering

- Internal sources: ERP, CRM, transaction systems. External sources: market research, govt data, social media, competitors.
- **Data quality dimensions:** Accuracy, Completeness, Consistency, Timeliness, Relevance, Accessibility, Validity.
- Completeness = no missing values; Timeliness = data is current/up to date; Consistency = same across systems.
- **Environmental scanning** = continuous monitoring of external factors (competitors, tech, regulation, economy).
- **Push delivery** = system sends info automatically (scheduled email/alert). **Pull delivery** = user requests info (ad-hoc query/login to dashboard).
- Methods: surveys, interviews, observation, existing records, web scraping.

3. Decision Making

- **Simon's 4 phases (very frequently asked):** Intelligence → Design → Choice → Implementation.
- Intelligence = identify problem; Design = develop alternatives; Choice = select best option; Implementation = execute + monitor.
- **Decision types (Gorry & Scott Morton):**
 - Structured = routine, repetitive, procedure known → automatable (e.g., reorder stock).
 - Semi-structured = partly standard, partly judgment (e.g., setting marketing budget) → needs DSS.
 - Unstructured = novel, no set procedure (e.g., entering new market) → needs human judgment.
- **Decision levels:** Strategic (top mgmt, long-term) → Tactical (middle mgmt) → Operational (day-to-day).
- **Rational model** = assumes full info, picks optimal solution.
- **Bounded rationality (Simon)** = limited info/time → leads to **satisficing** = picking “good enough,” not optimal.
- Decision styles: Directive (fast, task-focused), Analytical (data-heavy), Conceptual (creative, big-picture), Behavioral (people-focused).
- **GDSS** = Group Decision Support System, used for group decisions; risk in group decisions = groupthink.

4. Managing BI

- **BICC (BI Competency Center)** = cross-functional (business+IT) team governing BI strategy/standards/support — very common MCQ answer.
- **Centralized BI** = one team, consistent but slower.
- **Decentralized BI** = per-department, fast but inconsistent data definitions.
- **Federated/hybrid** = central standards + local execution (balance of both).
- **Data Steward** = owns data quality/definitions for a domain.
- **BI Sponsor** = provides funding/strategic direction (usually executive).
- BI maturity order (common sequence): Basic reporting → Data integration/DW → Self-service/data-driven culture → Predictive/prescriptive analytics → Pervasive BI.

5. BI User Segmentation

- **“Farmers vs Explorers/Miners”** (imp MCQ pair): Farmers = consume routine/standard reports repeatedly; Explorers/Miners = dig for new insight via ad-hoc analysis/data mining.
- Casual/consumer users → static dashboards & scheduled reports.
- Power users/analysts → self-service BI, OLAP, ad-hoc query tools.
- IT/BI developers → build & maintain BI infrastructure (ETL, admin).
- Executives need high-level KPIs/scorecards, not raw/granular data.
- Segmentation criteria: frequency of use, technical skill, need for ad-hoc vs standard reports, decision level.

6. Gathering BI Requirements

- Requirement types: Functional (what system does), Non-functional (performance/security/usability), Data requirements, Reporting requirements.
- **Elicitation techniques:** Interviews, Surveys, **JAD sessions** (Joint Application Design — group workshop with users+IT), Document/report analysis, Observation, **Prototyping** (mock-up to validate iteratively).
- **BRD** = Business Requirements Document — formal record of business needs.
- Key MCQ point: BI requirements are harder to gather than typical software requirements because needs are **exploratory** — users often “don’t know what they want until they see the data.”
- Requirements evolve throughout the BI project (not fixed like transactional systems).

7. Content and Knowledge Management

- **Tacit knowledge** = personal, experience-based, hard to document (e.g., a manager’s intuition).
- **Explicit knowledge** = documented, codified, easily shared (manuals, reports, SOPs).
- **SECI Model conversions (Nonaka) — very frequently asked:**
 - Socialization = Tacit → Tacit (mentoring, observation)
 - Externalization = Tacit → Explicit (documenting a best practice)
 - Combination = Explicit → Explicit (combining reports into new analysis)
 - Internalization = Explicit → Tacit (learning-by-doing until instinctive)
- KM lifecycle order: Create → Capture → Organize → Store → Share → Apply.
- **Content Management (CM)** manages documents/files; **Knowledge Management (KM)** manages human expertise/know-how — commonly tested distinction.
- KM tools: knowledge repositories, communities of practice, expert locators, wikis/portals.

8-11. Strategic BI, Visual Analytics, Data Representation & Data Collection/Binding

(Low weightage — typically only 4-5 MCQs total across these 4 topics, so just the highest-yield one-liners below.)

- **BI strategy** = aligning BI investments with business goals; can be **Top-down** (leadership-driven) or **Bottom-up** (department/data-driven).

- **Balanced Scorecard** (if strategy is asked) has 4 perspectives: **Financial, Customer, Internal Process, Learning & Growth**.
 - **CSF vs KPI**: CSF = area that must go right; KPI = metric measuring progress toward it.
 - **Visual analytics** = automated analysis + **interactive** visualization; plain **information visualization** can be static — this “interactive vs static” distinction is the #1 tested point.
 - Why visuals matter: humans process visual/graphical info **faster** than raw numbers/text (pre-attentive processing) → faster insight, pattern/anomaly detection.
 - **Chart-to-data matching** (common MCQ format): Bar = category comparison; Line = trend over time; Pie/Treemap = part-to-whole; Scatter = relationship/correlation; Map = geographic; Histogram/Box plot = distribution; Heat map = intensity/density.
 - **Data-ink ratio (Tufte)**: maximize data-representing ink, minimize decorative “chartjunk.” Pie charts avoided for >5-6 categories (use bar instead).
 - **3 data types (not 2) — watch for this trap**: Structured (fixed schema, SQL/Excel), Semi-structured (XML, JSON — has tags but no rigid schema), Unstructured (text, images, video, social media — no schema).
 - Commonly cited stat: **~80-90% of enterprise data is unstructured.**
 - **Data binding** = linking a data source to a visual/dashboard so it updates with data; **Static binding** = fixed snapshot; **Dynamic/live binding** = auto-updates.
 - **Primary data** (collected firsthand: survey/interview) vs **Secondary data** (already existing: reports/published data).
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High-Yield Cross-Topic MCQ Traps

- DIKW is 4 levels, not 3 (Data, Information, Knowledge, Wisdom) — don’t forget Wisdom.
- Analytics types are 4, in strict order: Descriptive, Diagnostic, Predictive, Prescriptive.
- Data types are 3, not 2: Structured, Semi-structured, Unstructured.
- Simon’s model is 4 phases (some sources add a 5th “Monitoring/Review”).
- SECI is exactly 4 conversions — know direction of each (which is Tacit→Explicit vs Explicit→Tacit).
- BICC = business + IT combined team, not IT-only.
- Farmers = routine/standard consumers; Explorers/Miners = ad-hoc/new insight seekers (don’t swap these).
- OLAP ≠ OLTP: OLAP = analysis; OLTP = transactions.
- CM (content) ≠ KM (knowledge): CM = documents; KM = human expertise.
- Visual analytics = interactive; Information visualization can be static — visual analytics is the broader/more advanced term.

MS Excel MCQ Quick Revision

MS Excel — MCQ Quick Revision (High Weightage)

1. Functions

- **Function** = predefined built-in formula (e.g., SUM, IF); **Formula** = any expression you write starting with = (may or may not use a function) — common conceptual MCQ.
- **Logical:** IF, AND, OR, NOT, IFERROR (traps/handles errors), IFS (multiple conditions without nesting).
- **Statistical:** AVERAGE, MEDIAN, MODE, STDEV, VAR, COUNT (counts numbers only), COUNTA (counts non-blank incl. text), COUNTBLANK.
- **Conditional counting/summing (imp trap):** COUNTIF/SUMIF = single condition; COUNTIFS/SUMIFS = multiple conditions (AND logic).
- **Text functions:** LEFT, RIGHT, MID (extract characters), CONCATENATE/CONCAT/& (join text), TRIM (removes extra spaces), LEN (character count), UPPER/LOWER/PROPER.
- **Date/Time:** TODAY() (no args, current date), NOW() (date+time), DATEDIF (difference between two dates), YEAR/MONTH/DAY.
- **Math:** SUM, ROUND (rounds to n decimals), ROUNDDOWN/ROUNDUP, ABS (absolute value), SQRT.
- **Nested IF** = IF inside another IF for multiple conditions; can get complex — IFS/SWITCH are cleaner alternatives (often the “better practice” MCQ answer).
- **Array formula** = entered with Ctrl+Shift+Enter (shows as {=...}); operates on multiple values at once — distinct from a normal formula.

2. Formula

- Every formula **must start with =** (sometimes + or - also works but = is standard).
- **Reference types (very frequently tested):**
 - **Relative reference** (A1) — changes when copied to another cell.
 - **Absolute reference** (\$A\$1) — stays fixed when copied.
 - **Mixed reference** (\$A1 or A\$1) — only column or only row is fixed.
 - **F4 key** toggles between reference types while editing a formula.
- **Order of operations** in Excel follows standard math (BODMAS/PEMDAS): Brackets → Exponents → Mult/Div → Add/Sub.
- **Common error codes (high-frequency MCQ set):**
 - #DIV/0! → division by zero.
 - #N/A → value not available (classic in VLOOKUP when lookup value isn't found).
 - #REF! → invalid cell reference (often after deleting a referenced row/column).
 - #VALUE! → wrong data type used in formula (e.g., text where number expected).
 - #NAME? → Excel doesn't recognize text/function name (typo in function).
 - #NULL! → incorrect range operator (wrong use of space between two ranges).
- **Circular reference** = a formula that refers to its own cell (directly or indirectly) — Excel shows a warning.

3. Charts

- **Column chart** = vertical bars, compares values across categories.
- **Bar chart** = horizontal bars, same purpose as column but better for long category names.
- **Line chart** = shows trend over time/continuous data.

- **Pie chart** = part-to-whole, single data series only, best with few categories.
- **Scatter chart** = relationship/correlation between two numeric variables.
- **Combo chart** = combines two chart types (e.g., bar + line) to show two different metrics/scales together.
- **Sparkline** = tiny chart inside a single cell, used for quick trend view.
- Chart elements often tested: Chart Title, Axis Titles, Legend, Data Labels, Gridlines.
- You must **select the data range first**, then insert chart, for Excel to auto-detect series/categories.
- Pie charts are discouraged for many categories (readability) — bar/column preferred (same “best practice” trap as in visualization topic).

4. Pivots and Lookups

- **Pivot Table** = summarizes/aggregates large datasets by dragging fields into 4 areas: **Rows, Columns, Values, Filters**.
- Default aggregation in Values area: **SUM** for numeric fields, **COUNT** for text fields — common trap.
- Pivot Tables **do NOT auto-update** when source data changes — must **Refresh** manually (Right-click → Refresh, or Data → Refresh All).
- **Pivot Chart** = graphical view built from a Pivot Table.
- GETPIVOTDATA = function to pull a specific value out of a Pivot Table.
- **VLOOKUP** syntax: VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup]).
 - range_lookup = FALSE → exact match (safer, commonly tested as “should almost always use FALSE”).
 - range_lookup = TRUE or omitted → approximate match (default — frequent trap since it can silently give wrong results).
 - **VLOOKUP limitation:** can only search left-to-right (lookup column must be leftmost) — cannot look to its left.
- **HLOOKUP** = same concept but searches the **first row** horizontally instead of first column.
- **INDEX-MATCH** = more flexible combo; MATCH finds the position of a value, INDEX returns the value at that position; can look in **either direction** (overcomes VLOOKUP’s left-only limitation) — common “why use INDEX-MATCH over VLOOKUP” question.
- **XLOOKUP** (Excel 365) = modern replacement for VLOOKUP/HLOOKUP/INDEX-MATCH; searches both directions, defaults to **exact match**, can return arrays — frequently the “best/most modern option” MCQ answer.

5. Data Analysis ToolPak

5a. Descriptive Summaries

- **Analysis ToolPak** = an Excel **Add-in**; must be enabled via File → Options → Add-ins → Analysis ToolPak (not enabled by default) — common factual MCQ.
- Found under **Data tab → Data Analysis → Descriptive Statistics**.
- Output includes: **Mean, Standard Error, Median, Mode, Standard Deviation, Sample Variance, Kurtosis, Skewness, Range, Minimum, Maximum, Sum, Count.**
- **Mean** = average; **Median** = middle value (better for skewed data); **Mode** = most frequently occurring value.
- **Standard Deviation** = measure of spread/dispersion around the mean; **Variance** = square of standard deviation.
- **Skewness** = measures asymmetry of the distribution (positive skew = tail to the right).
- **Kurtosis** = measures “peakedness”/tail-heaviness of a distribution.
- **Range** = Maximum – Minimum.

5b. Correlation

- Found under **Data tab → Data Analysis → Correlation** (or CORREL() function for two variables directly).
- **Correlation coefficient (r)** ranges from **-1 to +1**.
 - +1 = perfect positive correlation; -1 = perfect negative correlation; 0 = no linear correlation.
- Correlation tool with multiple variables produces a **correlation matrix** (pairwise correlations).
- The measure used is the **Pearson correlation coefficient** by default in Excel’s tool/CORREL.
- **Key exam trap statement:** “Correlation does not imply causation” — a strong correlation doesn’t mean one variable causes the other.
- CORREL(array1, array2) = function form; gives same result as the ToolPak’s Correlation tool for two variables.

High-Yield Excel MCQ Traps

- VLOOKUP's 4th argument default is **TRUE (approximate match)**, not FALSE — a very common “gotcha” question.
- Pivot Table values default to SUM for numbers, COUNT for text — don't mix these up.
- COUNTIF/SUMIF = ONE condition; COUNTIFS/SUMIFS = MULTIPLE conditions — don't swap.
- \$A\$1 = absolute, A1 = relative, \$A1/A\$1 = mixed — F4 cycles through these.
- VLOOKUP can only look **rightward**; INDEX-MATCH and XLOOKUP can look **either direction**.
- Analysis ToolPak is an **add-in**, not enabled by default.
- Correlation range is **-1 to +1**; 0 means no linear relationship (not “no relationship at all”).
- Pivot tables need **manual refresh** — this is one of the most commonly tested “gotcha” facts.

Data Analytics Lifecycle MCQ Quick Revision

Data Analytics Lifecycle — MCQ Quick Revision

(Correct phase order is one of the most frequently tested facts — memorize the sequence below.)

Order: Discovery → Data Preparation → Model Planning → Model Building/Implementation → Quality Assurance → Documentation → Management Approval → Installation → Acceptance and Operation.

1. Discovery

- **First phase** — understand the business problem/objective before touching data.
- Identify stakeholders, business drivers, available resources (people, tech, data, time, budget).
- Team assesses whether it has the **skills, data, and tools** needed for the project.
- **Initial hypotheses** are framed here (not tested yet — just formed).
- Output: high-level project plan + defined business objective.

2. Data Preparation

- Involves **ETL, data cleaning, handling missing values, formatting/normalizing data** (“data conditioning”).
- Often includes creating an **analytics sandbox** — a separate workspace to explore/manipulate data without disturbing production systems (common MCQ term).
- **Exploratory Data Analysis (EDA)** — summary stats, visualization — happens in this phase.
- **High-yield stat:** commonly cited that this phase consumes **~60-80% of total project time/effort** — a frequently tested fact.

3. Model Planning

- Decide **which analytical method/technique** fits the business question and data type (regression, classification, clustering, etc.).
- Explore **relationships between variables** (correlation analysis often used here).
- Select variables to include in the model; determine the analytical workflow.
- Common tools referenced at this stage: R, SQL, SAS (for exploring data relationships before full model build).

4. Model Building / Implementation

- Actual **model is built and executed** using selected technique(s).
- Data split into **training and test datasets** (train-test split) — common MCQ point.
- Iterative: multiple models/parameters tried and compared.
- Tools: R, Python, SAS, WEKA, etc.

5. Quality Assurance

- Model outputs are **tested for accuracy, robustness, and reliability** before deployment.
- Validate performance metrics (accuracy, precision, recall, etc.) against **defined success criteria**.
- Testing done on **unseen/held-out data** to check generalization (not just training data).

6. Documentation

- Recording **methodology, assumptions, code, and findings** for reproducibility and future reference.
- Ensures **knowledge transfer** — someone else could reproduce/understand the work later.
- Includes both technical documentation and a business-facing summary of results.

7. Management Approval

- Results/model are **presented to stakeholders/decision-makers** for sign-off.
- Equivalent in purpose to “Communicate Results” in the standard 6-phase model (see mapping below).
- Focus: business value, ROI, key findings — decision on whether to proceed to deployment.

8. Installation

- Model is **deployed into the production environment**.
- Involves **integration with existing systems/infrastructure**.
- Corresponds to “Operationalize” in the standard 6-phase lifecycle model — common cross-reference MCQ.

9. Acceptance and Operation

- **Final phase** — model runs live in production.
- May include **User Acceptance Testing (UAT)**.
- Ongoing **monitoring, maintenance, and retraining** as data/business conditions change.
- The lifecycle is **iterative/cyclical, not strictly linear** — feedback can loop back to Discovery for the next cycle (frequently tested “true/false” statement).

Mapping to the Standard 6-Phase Data Analytics Lifecycle (EMC/Dell model — often cross-referenced in exams)

Standard 6-Phase Model	Maps to (this 9-phase version)
1. Discovery	Discovery
2. Data Preparation	Data Preparation
3. Model Planning	Model Planning
4. Model Building	Model Building/Implementation
5. Communicate Results	Quality Assurance + Documentation + Management Approval
6. Operationalize	Installation + Acceptance and Operation

High-Yield Traps

- The lifecycle is **iterative**, not one-way — teams often loop back to earlier phases (e.g., poor model results in QA can send you back to Data Preparation or Model Planning).
- **Data Preparation**, not Model Building, is the phase most often cited as consuming the **most time/effort** in a project.
- **Hypotheses are formed in Discovery**, not Model Planning — don’t confuse “forming hypotheses” (Discovery) with “selecting techniques to test them” (Model Planning).
- **Analytics sandbox** is created during **Data Preparation**, separate from the production environment — commonly tested term.
- **Train/test split** happens in **Model Building**, not Model Planning.
- **Management Approval** comes before **Installation** — sign-off is needed before deployment, not after.

- **Acceptance and Operation** is the only phase involving live production monitoring — QA (#5) tests before deployment, this phase (#9) monitors after deployment — don't mix these up.

Tableau and Data Analytics Tools MCQ Quick Revision

Tableau & Data Analytics Tools — MCQ Quick Revision (High Weightage)

1. Introduction to Tableau

- Tableau = drag-and-drop **data visualization/BI tool**, minimal coding required.
- Originally Tableau Software; **acquired by Salesforce (2019)** — commonly tested fact.
- **Tableau product family (frequently tested — know which is which):**
 - **Tableau Desktop** — build workbooks/visualizations.
 - **Tableau Server** — share dashboards **on-premise**.
 - **Tableau Online** — cloud-hosted sharing.
 - **Tableau Public** — free, but shared workbooks are **public**.
 - **Tableau Reader** — view-only, cannot edit.
 - **Tableau Prep** — data cleaning/preparation tool.
- Core terms: **Workbook** (file), **Worksheet** (single view/chart), **Dashboard** (combination of worksheets), **Story** (sequence of dashboards/worksheets for narrative).
- **Dimensions vs Measures (very high-frequency MCQ, color-coded):**
 - **Dimensions** = qualitative/categorical data, shown in **blue** (e.g., Region, Category).
 - **Measures** = quantitative/numeric data, shown in **green** (e.g., Sales, Profit).
- **Connection types:** **Live connection** = always queries the source in real time; **Extract** = snapshot saved locally in Tableau's **.hyper** format (faster performance, not live) — common trap: extract ≠ always up to date.

2. Intelligent Data Analysis (IDA)

- IDA = applying **AI/ML + statistical techniques** to automatically extract meaningful patterns from data (beyond manual/basic analysis).
- Techniques involved: classification, clustering, regression, association rule mining, pattern recognition.
- Closely related to **KDD (Knowledge Discovery in Databases) process** — order often tested: **Selection → Preprocessing → Transformation → Data Mining → Interpretation/Evaluation**.
- Goal: move from raw data to **actionable knowledge**, not just summary numbers.

3. Nature of Data

- **Qualitative (categorical) vs Quantitative (numerical)** data — top-level split.
 - Qualitative → **Nominal** (no order — e.g., gender, color) and **Ordinal** (ordered — e.g., rating: low/medium/high).
 - Quantitative → **Discrete** (countable — e.g., number of students) and **Continuous** (measurable — e.g., height, weight).
- **Levels of measurement (NOIR — frequently tested order):**
 1. **Nominal** — categories, no order (e.g., blood group).
 2. **Ordinal** — ordered categories, unequal/undefined gaps (e.g., satisfaction rating).
 3. **Interval** — ordered, equal intervals, **no true zero** (e.g., temperature in °C).
 4. **Ratio** — ordered, equal intervals, **has true zero** (e.g., weight, income, age).
- **Structured / Semi-structured / Unstructured** data (recap — 3 types, not 2).

- Data by time dimension: **Cross-sectional** (one point in time), **Time-series** (same variable over time), **Panel/Longitudinal** (multiple entities over time — combination of both).

4. Analytics Processes and Tools

- Typical analytics process flow: **Define problem → Collect data → Clean/prepare → Analyze → Visualize → Interpret → Act.**
- Tool categories (know which tool belongs to which category — common matching MCQ):
 - **Spreadsheet tools:** Excel.
 - **BI/Visualization tools:** Tableau, Power BI, Qlik.
 - **Statistical tools:** R, SPSS, SAS.
 - **Programming:** Python.
 - **Big data tools:** Hadoop, Spark.
- Tool selection depends on: data size/volume, complexity, user technical skill, need for real-time processing, and budget.

5. Analysis vs. Reporting (classic high-frequency MCQ pair)

- **Reporting** = presents **historical, “what happened”** data in a structured, often static format; usually **automated/recurring** (daily/weekly/monthly).
- **Analysis** = examines data to understand **“why it happened”** and **“what might happen next”**; more **exploratory/ad-hoc**, dynamic.
- Reporting = **output**; Analysis = the **process** leading to insight.
- Reporting is descriptive; Analysis can be diagnostic, predictive, or prescriptive (links to Topic 1 analytics types).

6. Modern Data Analytic Tools

- Categories: **BI/Visualization** (Tableau, Power BI, Qlik), **Programming/Statistical** (Python, R), **Big Data** (Hadoop, Spark), **Cloud Analytics** (AWS/Azure/GCP analytics services), **AutoML tools**.
- Growing trend: **self-service BI** — business users perform analysis without heavy IT/coding dependency.
- Cloud-based tools enable easier **collaboration and scalability** vs traditional desktop-only tools.

7. Data Sources in Tableau

- Tableau connects to: **Excel, CSV/text files, JSON, PDF, statistical files, relational databases** (SQL Server, MySQL, Oracle), **cloud databases** (Google BigQuery, Amazon Redshift, Snowflake), and **Web Data Connector**.
- **Live vs Extract connection** (recap from Topic 1) — key distinguishing factor for data sources.
- **Data Joining vs Data Blending (frequently confused pair):**
 - **Joining** = combines data **at row level within the same data source** (or same connection).
 - **Blending** = combines data from **multiple different data sources** at an **aggregate level** — used when a direct join isn't possible.
- The **Data Source page** in Tableau shows fields and lets you set up joins/relationships before analysis.

Self-Learning Topics — MCQ Quick Points

Tableau vs Power BI

- **Tableau:** stronger/more flexible custom visualizations, better for large/complex datasets, generally **higher cost**, uses **VizQL** engine.
- **Power BI:** Microsoft product, integrates tightly with **Excel/Office 365**, generally **cheaper (has a free tier)**, easier learning curve, uses **DAX** (Data Analysis Expressions) and **Power Query (M language)**.
- Power BI is often the “better choice” MCQ answer for organizations already using Microsoft ecosystem; Tableau often

the “better choice” for advanced/custom visual analytics.

Excel vs Tableau — when to use which

- **Use Excel when:** small-medium datasets, quick calculations, formula-heavy work, individual/ad-hoc analysis, pivot tables suffice.
- **Use Tableau when:** large datasets, need **interactive dashboards**, sharing insights with a wide audience, complex visual storytelling.

Emerging Analytics Tools — Python & R libraries

- **Matplotlib** — foundational/basic Python plotting library (most other libraries build on it).
- **Seaborn** — built on top of Matplotlib; better default styling, strong for **statistical visualizations**.
- **Plotly** — Python (and R) library for **interactive, web-based** visualizations.
- **ggplot2 (R)** — based on the **“Grammar of Graphics”** concept; layered/declarative approach to building plots — commonly tested distinguishing feature.

Cloud BI vs On-Premise

- **Cloud BI** (e.g., Tableau Online, Power BI Service): vendor-hosted, subscription/SaaS-based, accessible anywhere, scalable, minimal IT infrastructure needed, automatic updates.
- **On-Premise BI** (e.g., Tableau Server on-prem, Power BI Report Server): hosted on the organization’s own servers, more control/security/compliance, higher upfront infrastructure cost.
- Cloud = faster deployment; On-premise = better for sensitive/regulated data needing full control.

Real-Time Analytics vs Batch Processing

- **Real-time analytics:** processes data **as it arrives**; needed for fraud detection, stock trading, IoT monitoring; uses streaming tech (e.g., Kafka, Spark Streaming).
- **Batch processing:** processes data in **scheduled chunks/intervals** (e.g., nightly ETL jobs); suitable for historical reporting; less resource-intensive than real-time.

Which Tool for Which Task (quick-mapping table — common matching-style MCQ)

Task	Best-Fit Tool
Quick calculations, small dataset	Excel
Interactive dashboard for wide audience	Tableau / Power BI
Statistical/predictive modeling	R / Python
Big data processing (huge volume)	Hadoop / Spark
Real-time streaming analytics	Kafka / Spark Streaming
Deep integration with Microsoft/Office tools	Power BI
Complex, highly custom visualizations	Tableau

High-Yield Traps

- Dimensions = **blue** (categorical); Measures = **green** (numeric) — reversed color is a classic wrong-option trap.
- **Live connection** ≠ always current data — only **Live** is real-time; **Extract** is a saved snapshot.
- NOIR order is fixed: **Nominal → Ordinal → Interval → Ratio**; only Ratio has a **true zero**.
- **Joining** = same source, row-level; **Blending** = different sources, aggregate-level — don’t swap these.
- Reporting = “what happened” (descriptive, automated); Analysis = “why/what next” (exploratory) — a very common conceptual MCQ.
- Tableau was **acquired by Salesforce**, not Microsoft (Power BI is the Microsoft product) — easy trap if rushed.
- Power BI’s formula language is **DAX**; Tableau’s is **VizQL** — don’t mix up.
- Cloud BI = SaaS/subscription/vendor-hosted; On-premise = organization’s own infrastructure — cost/control trade-off is the recurring theme tested.

Visualization Algorithms, Encodings and Chart Taxonomy MCQ Quick Revision

Visualization Algorithms, Encodings & Chart Taxonomy — MCQ Quick Revision

1. Visualization Algorithms

- **Visualization pipeline (frequently tested order):** Raw Data → Data Transformation/Filtering → Visual Mapping → View Transformation → Rendering.
- **Force-directed layout** — physics-based algorithm for network/graph visualizations; nodes repel each other, connected nodes are pulled together by edges (springs) — used in network diagrams.
- **Treemap algorithm** — space-filling, hierarchical; rectangle size represents value; **“squarified” treemap** keeps rectangles closer to square shape for readability.
- **Binning algorithm** — groups continuous data into intervals/bins; used specifically in **histograms**.
- **Dimensionality reduction algorithms** (PCA, t-SNE) — used to visualize high-dimensional data in 2D/3D — common in advanced analytics visualization.
- **Clustering algorithms** (e.g., k-means) sometimes applied before visualization to group/color-code similar data points.
- **Radial/circular layout** — alternative to linear layout, often used for hierarchical or cyclical data.

2. Visual Encodings

- **Visual encoding** = mapping a data attribute to a visual property (position, size, color, shape, etc.).
- **Cleveland & McGill hierarchy of perceptual accuracy (HIGHLY tested — memorize this order, most → least accurate):**
 1. **Position** (along a common/aligned scale) — most accurate
 2. **Position** (along non-aligned scale)
 3. **Length**
 4. **Angle / Slope**
 5. **Area**
 6. **Volume**
 7. **Color (hue/saturation/density)** — least accurate for precise value judgment
- **Color** — encodes **categorical data via hue** (different colors = different categories) or **quantitative data via saturation/luminance** (darker/lighter = higher/lower value). Color schemes: **Sequential** (single hue, light→dark, for ordered data), **Diverging** (two hues from a midpoint, for data with a meaningful center like positive/negative), **Qualitative** (distinct hues, for categories with no order).
- **Size** — encodes quantitative values (e.g., bubble size); humans are **less accurate** at comparing area/size differences than length/position — a common exam point.
- **Shape** — encodes categorical data (different markers per category); effective only for a **small number of categories (~5-7 max)** — too many shapes reduce readability.
- **Lines** — encode connections, trends, or relationships between points (e.g., line charts, network edges).
- **Axes** — define the reference scale for position encoding. **Bar chart axes must start at zero** — a truncated/non-zero axis is a classic “misleading chart” trap question.
- **Scaling** — **Linear scale** = equal visual distance for equal value difference; **Logarithmic scale** = used when data spans multiple orders of magnitude (compresses large ranges) — common “when to use log scale” MCQ.
- **Annotation** — text labels/callouts added to highlight specific points or insights; improves interpretability but

excessive annotation causes clutter (“chartjunk” — recap from data-ink ratio concept).

3. Taxonomy of Data Visualization

A. Comparison Charts

- **Bar chart** — compares categories using bar **length**; types: simple, **grouped/side-by-side** (compares subcategories next to each other), **stacked** (shows part-to-whole within each category).
- **Box plot** — shows distribution via the **5-number summary**: Minimum, Q1, Median, Q3, Maximum; highlights **outliers** — commonly tested for identifying spread/skew.
- **Histogram** — shows frequency distribution of **continuous** data using **bins**; **key distinguishing trap vs bar chart**: histograms have **no gaps** between bars (continuous data) while bar charts have gaps (discrete categories).
- **Gantt chart** — shows project schedule/timeline; tasks (rows) vs time (x-axis); used in **project management**.
- **Bullet graph** — compact chart comparing an actual measure against a **target**, with qualitative performance ranges (e.g., poor/satisfactory/good) in the background; created by **Stephen Few** as an alternative to gauge/speedometer charts — frequently tested “who invented it” fact.
- **Side-by-side bar chart** — places bars for multiple series next to each other within each category for direct comparison.

B. Tables

- **Text table** — simple grid of exact values, no visual/color encoding; best for **precise lookups**, not pattern-spotting.
- **Highlight table** — a text table with **color-coded cells** (like a heat map) layered on top of exact values — combines precision (numbers) with pattern visibility (color).

C. Hierarchies and Relationships

- **Pie chart** — part-to-whole, **single data series only**; best with **few slices ($\leq 5-6$)** — too many slices hurt readability (recurring “best practice” trap).
- **Stacked bar chart** — shows part-to-whole composition **within** each bar while still allowing comparison **across** categories.
- **Treemap** — hierarchical, **space-filling rectangles**; rectangle **size = value**; nested rectangles represent hierarchy levels — good for showing part-to-whole across many categories where a pie chart would fail.

D. Changes Over Time

- **Line chart** — the standard chart for showing **trend over continuous time**.
- **Dual-line chart** — compares two line series together; if using **dual axes** (two different y-axis scales), this is a **common “misleading chart” trap** since it can visually imply false correlation.
- **Area chart** — like a line chart but the area beneath the line is filled, emphasizing **magnitude/volume** over time; can be **stacked** to show part-to-whole trends over time.

E. Connections and Relationships

- **Scatter plot** — shows **relationship/correlation** between two continuous variables (recap from earlier correlation topic).
- **Symbol map** — geographic locations marked with symbols, often sized/colored by a value.
- **Heat map** — uses **color intensity** to represent value magnitude across a matrix or geographic region (**choropleth map** = geographic version of heat map).
- **Packed bubble chart** — circles sized by value, grouped together in a compact, non-hierarchical layout; used for part-to-whole/comparison in limited space (different from treemap which shows explicit hierarchy).

High-Yield Traps

- **Cleveland-McGill order** is the single most tested fact here: **Position > Length > Angle/Slope > Area > Volume > Color** (best to worst for accurate perception).
- **Histogram vs Bar chart**: Histogram = continuous data, binned, **no gaps**; Bar chart = categorical/discrete data, **has gaps** between bars.

- **Bar chart axis must start at 0** — starting elsewhere exaggerates differences (classic “misleading visualization” question).
- **Bullet graph** was created as a better alternative to the **gauge/speedometer chart** — commonly paired in MCQs.
- **Treemap vs Stacked Bar:** both show part-to-whole/hierarchy, but Treemap uses **area/nested rectangles**, Stacked Bar uses **segmented length** — know which is which when a diagram is shown.
- **Dual-axis line charts** can visually mislead by implying correlation between unrelated scales — treat as a red flag in “identify the misleading chart” questions.
- **Color is the LEAST accurate encoding** for precise quantitative comparison (per Cleveland-McGill) — yet it’s the most common encoding for **categorical** distinction (hue) — don’t confuse “accurate for values” vs “useful for categories.”
- **Shape encoding** should be limited to **~5-7 categories max** — same logic as pie chart slice limits.
- **Pie chart:** single series + few categories; **Stacked bar/Treemap:** better for many categories or multiple series showing hierarchy.

Tableau Calculations, Filters, Interactivity and Dashboard Design MCQ Quick Revision

Tableau Calculations, Filters, Interactivity & Dashboard Design — MCQ Quick Revision

1. Choosing Appropriate Visuals

- Chart choice should match the **question being answered**, not personal preference:
 - Comparison → Bar chart
 - Distribution → Histogram, Box plot
 - Relationship/correlation → Scatter plot
 - Composition/part-to-whole → Pie chart, Treemap, Stacked bar
 - Trend over time → Line chart, Area chart
 - Geographic → Map, Symbol map, Heat map
- Match **number of variables** to chart type: 1 variable → bar/histogram; 2 variables → scatter/line; part-to-whole → pie/treemap/stacked bar.
- Avoid misleading types: **3D charts** (distort perception), **pie charts with many slices**, **dual-axis charts with mismatched scales** — recurring “avoid this chart” MCQ theme.

2. Applying Calculations Using Functions & Statistics

A. Numeric Calculations

- Basic operators: +, -, *, /.
- Aggregate functions: SUM, AVG, MIN, MAX, COUNT (counts all), **COUNTD** (counts **distinct** values — common trap vs COUNT).
- Other common functions: ROUND, ABS, SQRT, POWER.

B. String Calculations

- LEFT, RIGHT, MID — extract characters from a string.
- LEN — string length; UPPER/LOWER — case conversion; TRIM — removes leading/trailing spaces.
- REPLACE — substitutes part of a string; CONTAINS — checks if a substring exists (returns Boolean); FIND — returns position of a substring; SPLIT — breaks a string into parts using a separator.
- String concatenation uses the **+** operator (or CONCAT in newer versions) — different from Excel’s &.

C. Date Calculations

- DATEDIFF('unit', date1, date2) — difference between two dates in specified unit (day/month/year, etc.) — very commonly tested syntax.
- DATEADD('unit', number, date) — adds/subtracts time from a date.
- **DATEPART vs DATENAME (frequently confused pair):** DATEPART returns the date component as a **number** (e.g., month = 7); DATENAME returns it as a **string/name** (e.g., month = “July”).
- TODAY() — current date; NOW() — current date + time.

D. LOD (Level of Detail) Expressions — HIGH-YIELD TOPIC

- LOD expressions compute values at a **different granularity than the view** — the single most tested Tableau

calculation concept.

- **Three types (know the syntax and behavior of each):**
 - **FIXED** — {FIXED [Dimension] : AGG(Measure)} — computes using **only the specified dimensions**, ignoring the view's other dimensions/filters (mostly).
 - **INCLUDE** — {INCLUDE [Dimension] : AGG(Measure)} — computes at a **finer** level of detail than the view (adds extra dimensions to the calculation).
 - **EXCLUDE** — {EXCLUDE [Dimension] : AGG(Measure)} — computes at a **coarser** level of detail than the view (removes a dimension from the calculation).
- **Classic use case:** FIXED for a customer-level total that stays constant regardless of view-level date filters.
- **Order of operations trap (very frequently tested — memorize this sequence):** Extract Filters → Data Source Filters → **Context Filters** → **FIXED LOD Expressions** → Dimension Filters → **INCLUDE/EXCLUDE LOD Expressions** → Measure Filters → Table Calculation Filters.
 - Key exam point: **FIXED LOD is calculated before dimension filters are applied**, so a normal dimension filter will NOT affect a FIXED calculation unless it's set as a **Context Filter** first.

3. Data Sorting & Filters

- **Sorting** — ascending/descending; can sort by the field itself or by an associated measure (e.g., sort category by total sales).
- **Filter order of operations (high-frequency MCQ — same sequence as above, tested standalone too):**
 1. Extract Filters
 2. Data Source Filters
 3. **Context Filters**
 4. Dimension Filters
 5. Measure Filters
 6. Table Calculation Filters
- **Context Filter** — an independent filter that all subsequent filters are based on; commonly used to **improve performance** on large datasets by first reducing data to a smaller subset, and also required to make dimension filters affect FIXED LOD calculations.
- **Top N filter** — a dimension filter type (e.g., “Top 10 customers by Sales”) — usually needs to be set up correctly relative to context filters to work as expected.
- **Quick Filters** — interactive filter controls shown on a worksheet/dashboard for end users.

4. Interactive Visualization

A. Data Updating

- **Live connection** — automatically reflects changes in the underlying data source (always current).
- **Extract** — a saved snapshot (.hyper file); does **NOT auto-update**; requires a **manual or scheduled refresh** (Tableau Server can automate this on a schedule) — common trap: assuming extracts are always live.

B. Visual Updating (Dashboard Actions — high-frequency topic)

- **Filter action** — clicking/selecting a mark in one view filters another view.
- **Highlight action** — selecting a mark highlights related marks elsewhere (doesn't filter, just visually emphasizes).
- **URL action** — clicking a mark opens a specified URL (e.g., links to external page with more detail).
- **Parameter action** — selecting a mark updates a parameter's value dynamically.
- **Set action** — selecting marks adds/removes them from a Set, which can then drive calculations elsewhere.
- **Parameters** — user input controls letting viewers dynamically change a calculation or view (e.g., toggle between metrics).
- **Tooltips** — show additional detail on hover; can be customized to include mini-visualizations.

5. Dashboard Design

- **Dashboard** = combination of multiple worksheets/objects into a single consolidated view.

- **Layout containers:** **Horizontal** and **Vertical** containers organize objects for consistent alignment/resizing — commonly tested as the correct way to structure a responsive dashboard.
 - **Dashboard objects:** Worksheets, Text, Images, Web Page, Blank space, Containers.
 - **Sizing options:** **Fixed size** (specific pixel dimensions) vs **Automatic/Range size** (adjusts to device/window) — device-specific layouts (Desktop/Tablet/Phone) can be designed separately.
 - **Story** — a sequence of dashboards/worksheets arranged to build a narrative (recap from Topic 1's Tableau terms).
 - Best practices commonly tested: limit number of visuals per dashboard (avoid clutter), maintain consistent color scheme across views, ensure dashboard actions are used for interactivity rather than static duplication.
-

High-Yield Traps

- **LOD and Filter order of operations is the #1 tested fact in this section** — Context Filters apply **before** FIXED LOD, which applies **before** normal Dimension Filters.
- **FIXED** ignores view dimensions (uses only what's specified); **INCLUDE** adds detail; **EXCLUDE** removes detail — don't mix up direction (finer vs coarser).
- **COUNT vs COUNTD:** COUNT counts all non-null rows; COUNTD counts only **distinct** values — common numeric-function trap.
- **DATEPART (number) vs DATENAME (string)** — reversed in wrong-answer options frequently.
- **Context Filter's dual purpose:** (1) improves performance on large data, (2) is the mechanism required for regular filters to affect FIXED LOD results — both facts get tested.
- **Extract ≠ live data** — must be refreshed manually or on a schedule; a common "which statement is true" trap.
- **Filter action vs Highlight action:** Filter actually **removes** unrelated marks from view; Highlight only **visually emphasizes** without removing data — frequently swapped in MCQs.
- **Horizontal/Vertical containers**, not free-floating placement, are the recommended way to build a maintainable, responsive dashboard layout.